

Experiment-1: Create a New Process and Display
PID and PPID

pseudo code:

```
Start
call fork() to create a child process
If fork() Returns 0
    This is child process
    Get PID using getpid()
    Get PPID using getppid()
    Display child PID and PPID
Else If fork() > 0
    This is parent process
    Get PID using getpid()
    Get PPID using getppid()
    Display parent PID and PPID
Else
    Display "fork failed"
Stop
```

sample output:

Parent process

PID : 4521

PPID : 3312

Child process

PID : 4522

PPID : 4521

Experiment - 2: Copy contents of one file to another using system call

pseudo code:

Start

Open source file in Read mode

Open destination file in write mode

while source file is not empty

Read data

write data into destination file

Close both files

Display "File copied successfully"

Stop

Sample output:

Source file : input.txt

Destination : output.txt

File copied successfully

Experiment - 3: CPU Scheduling - FCFS

Pseudo code:

Start

Input number of processes

For each process

Input Burst time

waiting time of the first process = 0

For Remaining processes

waiting Time = sum of previous burst time

Turnaround time = waiting Time + Burst time

Calculate Average waiting Time

Calculate Average Turnaround Time

Display Results

stop

Sample output:

Process	BT	WT	TAT
		0	5
P ₁	5		8
P ₂	3	5	
P ₃	2	8	10

Average waiting Time = 4.33

Average Turnaround Time = 7.67

Experiment 04: CPU scheduling - shortest Job first (SJF)

Pseudo code:

Start

Input number of processes

Input Burst time

Sort processes by Burst time

Waiting Time at First process = 0

For Remaining processes

waiting time = sum of previous burst times

Turn around time = waiting Time + Burst time

Calculate averages

Display Results

Stop

Sample output:

Process	BT	WT	TAT
P ₃	2	0	2
P ₂	3	2	5
P ₁	5	5	10

Average waiting time = 2.33

Average Turnaround time = 5.67

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Experiment 05: Priority scheduling (non-preemptive)

Pseudo Code:

Start

Input number of processes

Input Burst Time

Input priority

Sort by priority

waiting Time of first process = 0

calculating waiting time

calculate Turnaround Time

calculate averages

Display Results

Stop

sample output:

Process	BT	priority	WT	AT
			0	3
P ₂	3	1	3	8
P ₁	5	2	8	12
P ₃	4	3		

Average waiting Time = 3.67

Average turnaround Time = 7.64

Experiment - 06: pre-emptive priority scheduling

Pseudo code:

```

Start
Input processes
Input Arrival time
Input Burst Time
Input priority
Repeat until all processes finish
Select highest priority process
Execute for one unit
If higher priority process arrives
    switch process
update remaining Time
calculate waiting Time
calculate Turnaround time
Display Results
Stop
    
```

sample output:

Process	AT	BT	priority	WT	TAT
P ₁	0	5	2	5	10
P ₂	1	3	1	0	3
P ₃	2	4	3	6	10

Avg. waiting Time = 3.67

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Experiment - 07: Non-preemptive SJF

pseudocode:

Start
Input processes
Input arrival time
Input Burst Time
Repeat until all process finish
Choose shortest burst among arrived processes
Execute completely
Calculate waiting time
Calculate Turnaround Time
Display Results
Stop

sample output:

Process	AT	BT	WT	TAT
			0	6
P ₁	0	6	4	7
P ₂	2	3	2	4
P ₃	4	2		

Average waiting Time = 2.00

Average Turnaround Time = 5.67

Experiment - 08: Round Robin scheduling

Pseudo code:

Start

Input number of processes

Input Burst time

Input time Quantum

while processes Remain

Execute each process for time Quantum

If process is unfinished

Put in back in ready Queue

Else mark completed

Calculate waiting time

Calculate Turnaround time

Display Results

Stop

Sample output:

Time Quantum = 2

Process	BT	WT	TAT
P ₁	5	7	12
P ₂	3	6	9
P ₃	4	7	11

$$\text{Avg. WT} = 6.67$$

$$\text{Avg. TAT} = 10.67$$

Verified